

Vicent Kamya

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Profile

Ph.D. chemist specializing in organic and process chemistry with hands-on experience in multistep synthesis, reaction optimization, and scalable continuous-flow production of pharmaceutically relevant compounds. Demonstrated ability to translate laboratory chemistry into robust, reproducible, and automation-enabled processes. Experienced in pharmaceutical chemical development and process development, with strong focus on reaction scalability, process robustness, data integrity, and safe laboratory practices.

Education

University of Connecticut

Ph.D. in Chemistry (GPA 3.8)

Concentration: Organic Chemistry

Storrs, CT

05/2027

- Designed, optimized, and scaled a continuous-flow pharmaceutical process for the conversion of dihydroartemisinic acid to artemisinin, achieving 67–70% yield, >95% purity, and >97% conversion. Focus was laid on reproducibility, safety and efficiency.
- Built and validated a bench-scale flow production unit integrating photochemical and thermal steps, emphasizing reaction scalability, throughput, and process robustness.
- Developed novel artemisinin derivatives (disulfide-linked ditriazole dimers) via multistep organic synthesis.
- Engineered automated flow-chemistry platforms using Raspberry Pi, Pico W, PCA9685, sensors, MFCs, and Python for process control, calibration, and data logging.
- Maintained detailed process documentation and experimental records supporting data integrity and reproducibility.
- Co-ordinated and enforced lab safety through waste management and overseeing cleaning activities

University of Nairobi

M.Sc. in Chemistry (GPA 3.9)

Concentration: Organic Chemistry

Nairobi, Kenya

09/2018-09/2022

- Conducted phytochemical analysis of Pentas species targeting leishmanicidal activity.
- Isolated and characterized natural products using chromatography, NMR, and LC–MS.
- Performed structure–activity relationship and computational docking studies.
- Advanced analytical work conducted at Uppsala University, Sweden.

Kyambogo University

B.Sc. In Education (GPA 4.34/5.0)

Concentration: Chemistry and Mathematics

Kampala, Uganda

08/2013-12/2016

- Final Project: Analysis of heavy metals in soil and water samples around school laboratories

Work Experience

University of Connecticut

Graduate Assistant

Storrs, CT

09/2022-01/2026

- Designed and optimized multistep organic reactions with emphasis on yield, purity, reproducibility, and scalability. Collaborated with a physical chemistry team to synthesize quantum dots using our flow platform with quantum dots exceeding 60%, much better than the batch equivalent (<30%).
- Designed and implemented continuous-flow and semi-automated platforms for pharmaceutical intermediate synthesis. Focused on production of artemisinin and its derivatives
- Developed Python-based process control algorithms for reagent delivery, flow correction, and calibration.
- Conducted analytical characterization using HPLC, LC-MS, NMR, IR, and UV-Vis.
- Worked in a collaborative, cross-functional research environment, communicating results to support process decision-making.
- Mentored undergraduate researchers in laboratory technique, safety and synthesis workflows.

Busitema University - Natural Products Research & Innovation Center

Research Assistant

Mbale, Uganda

09/2020-01/2021

- Developed extraction and purification workflows for bioactive phytochemicals (natural products)
- Characterized compounds via NMR and LC-MS, correlating structure with biological activity.
- Contributed to discovery efforts for natural-product-derived therapeutics.

Volunteer Experience

International Christian Fellowship (ICF)

President

Storrs, CT

05/2024-01/2026

- Presided over all executive and membership meetings
- Represented ICF at official functions and activities
- Co-ordinated tasks, events and prayer meetings
- Visionary roles; dictating the direction of the organization, consolidating relationships with volunteers
- Oversaw the actions and duties of all other officers

Technical Skills

Synthetic & Process Chemistry: Multistep organic synthesis, pharmaceutical process development, reaction scalability, throughput optimization, process robustness, continuous-flow chemistry, photochemistry

Analytical Techniques: HPLC, LC-MS, NMR, IR, UV-Vis, chromatography.

Automation & Instrumentation: Flow-reactor design, peristaltic pumps, sensors, MFCs, Raspberry Pi, Pico W, PCA9685.

Software, Programming & Data: Python, experimental automation, data analysis, Reaxys, Sci-Finder.

Soft Skills

Process-oriented thinking, scientific documentation, cross-functional collaboration, problem solving, attention to detail, excellent technical communication

Languages

English (Fluent), Luganda, Lusoga, Swahili